Pearson Edexcel International A Level · Pure Mathematics 1

Chapter 8 — Differentiation
Original Practice Questions

30 complete questions · Questions followed by answers and suggested mark schemes

- Answer all parts and show sufficient working.
- Give exact answers unless a degree of accuracy is requested.
- Angles are in radians unless degrees are explicitly indicated.
- Diagrams are not necessarily drawn to scale.
- The marks for each subpart are shown in brackets.

P1 第8章 · 微分
英文题目 · 中文解析 · 建议评分细则
本册30题；P1、P2、P3全套25章共750题。

内部复核稿，未完成外部教师逐题审阅。
Original practice material. Not a Pearson examination paper.

1. P1-C08-R01 · P1 · Chapter 8

The curve C has equation $y = 2x^3 - 5x^2 + 4x - 7$.

(a) Find $\dfrac{dy}{dx}$ and $\dfrac{d^2y}{dx^2}$. (4)

(b) Find the gradient of C at the point where $x = -1$. (2)

(Total for Question 1 is 6 marks)

C is the curve $y = x^2 + 3x - 2$. Point P on C has $x = 2$.

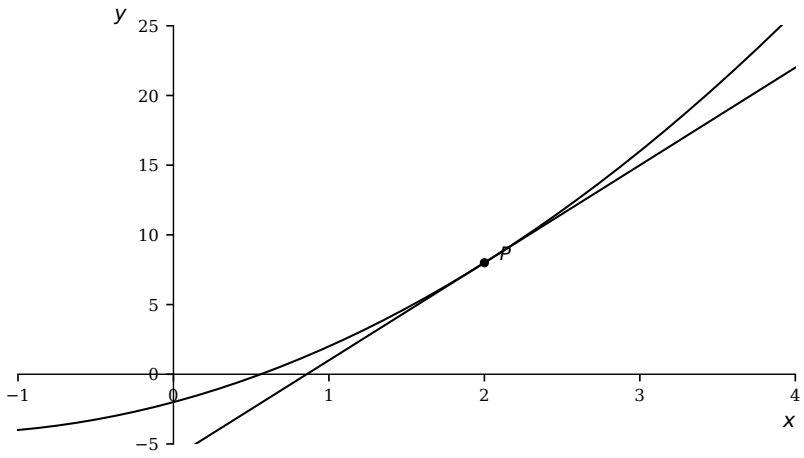


Figure 2

(a) Find an equation of the tangent to C at P. (3)

(b) Find where this tangent meets the x-axis and the y-axis. (2)

(Total for Question 2 is 5 marks)

3.

The curve C has equation $y = 4\sqrt{x} + \dfrac{3}{x}$ for $x > 0$.

(a) Find dy/dx, using powers of x. (2)

(b) Find the equation of the normal at the point where $x = 1$. (3)

(Total for Question 3 is 5 marks)

4.

A curve has equation $y = ax^3 + bx^2 + 1$. At the point $(1, 4)$, its tangent has gradient 5.

(a) Form two equations in a, b and find a, b. (4)

(b) Find the value of d^2y/dx^2 at $x = 2$. (2)

(Total for Question 4 is 6 marks)

$y = \dfrac{3x^3 - 2x + 5}{x}$, where $x \neq 0$.

(a) Express y as a sum of powers of x and hence find dy/dx. (3)

(b) Find the gradient at $x = -1$ and the value of y there. (2)

(Total for Question 5 is 5 marks)

6.

The curve C is $y = x^3 - 6x + 2$.

(a) Find the exact x-coordinates of the points where the tangent is parallel to the x-axis. (3)

(b) Find the exact y-coordinates of these points. (2)

(Total for Question 6 is 5 marks)

7.

At the point $P = (2, 5)$ on a curve, the gradient is $-\frac{1}{2}$.

(a) Find equations of the tangent and normal at P. (4)

(b) Find the area of the triangle formed by the tangent and the coordinate axes. (3)

(Total for Question 7 is 7 marks)

8. P1-C08-R08 · P1 · Chapter 8

C has equation $y = (x - 2)(x^2 + x + 1)$.

(a) Expand the equation and find dy/dx. (3)

(b) Find all points on C where the tangent has gradient 7. (3)

(Total for Question 8 is 6 marks)

A road profile is modelled by $h = 0.02x^2 + 0.3x + 4$ metres for $0 \leq x \leq 20$, where x is horizontal distance in metres.

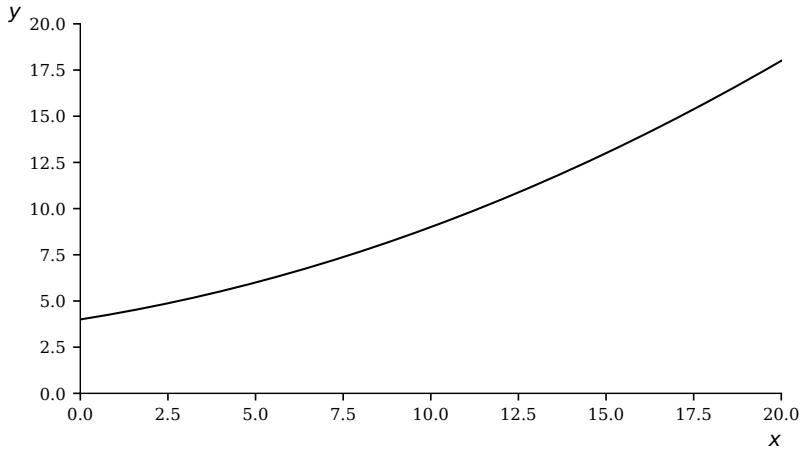


Figure 9

(a) Find dh/dx and its value at $x = 10$. (3)

(b) The gradient must not exceed 0.9. Find the part of the modelled road satisfying this condition. (3)

(Total for Question 9 is 6 marks)

The curve C has equation $y = 5x^{3/2} - 2x^{1/2}$ for $x > 0$.

(a) Find dy/dx. (2)

(b) At the point where $x = 4$, find

(i) the coordinates of the point and the tangent gradient, (3)

(ii) an equation of the tangent. (2)

(Total for Question 10 is 7 marks)

A curve is $y = ax^2 + bx + c$. Its tangent at $(0, 3)$ is $y = -4x + 3$, and the curve passes through $(2, 7)$.

(a) Find a, b, c. (4)

(b) Find where the curve has a tangent parallel to $y = 8x + 1$. (3)

(Total for Question 11 is 7 marks)

C is the curve $y = \dfrac{8}{x^2} + x$, with $x \neq 0$.

(a) Find dy/dx and d^2y/dx^2. (4)

(b) Find the equation of the tangent at $x = -2$. (3)

(Total for Question 12 is 7 marks)

13. P1-C08-R13 · P1 · Chapter 8

The curve C is $y = x^2 - 6x + 10$. A normal to C is parallel to $y = \frac{1}{2}x + 4$.

(a) Find the point of contact of this normal. (3)

(b) Find the equation of the normal and its x-intercept. (3)

(Total for Question 13 is 6 marks)

$f(x) = 3x^4 - 8x^3 + 2x.$

(a) Find $f''(x)$. (3)

(b) Find the values of x for which $f''(x) = 0$, and evaluate $f'(x)$ at each. (4)

(Total for Question 14 is 7 marks)

15. P1-C08-R15 · P1 · Chapter 8

The curve C is $y = x^3 + px + q$. At $P = (1, 2)$, its tangent is parallel to $y = 5x - 7$.

(a) Find p and q. (3)

(b) Find the equation of the normal at P. (3)

(Total for Question 15 is 6 marks)

16. P1-C08-R16 · P1 · Chapter 8

C is the curve $y = (2x + 1)^2$.

(a) By expanding first, find dy/dx. (3)

(b) At what point on C is the tangent perpendicular to the line $2x + 8y = 3$? (3)

(Total for Question 16 is 6 marks)

At a point P on $y = x^3 - 2x^2 + 1$, the tangent has gradient -1.

(a) Find all possible x-coordinates of P. (3)

(b) For $x = 1$, find the normal and its intersection with the y-axis. (4)

(Total for Question 17 is 7 marks)

18. P1-C08-R18 · P1 · Chapter 8

A curve is $y = 6x^{-1/2} + 2x$, for $x > 0$.

(a) Find the derivative and its value at $x = 9$. (3)

(b) Find the equation of the tangent at $x = 9$. (3)

(Total for Question 18 is 6 marks)

19. P1-C08-R19 · P1 · Chapter 8

The tangent to $C : y = x^2 + 2x$ at P passes through the origin. Let the x-coordinate of P be a.

(a) Show that $a = 0$. (3)

(b) Find the equation of this tangent and its perpendicular line through $(2, 1)$. (3)

(Total for Question 19 is 6 marks)

C is $y = 2x^3 - 3x^2 + 1$.

(a) Find the equations of the tangents at the points with $x = 0$ and $x = 1$. (3)

(b) Find the distance between these two tangents. (2)

(Total for Question 20 is 5 marks)

21. P1-C08-R21 · P1 · Chapter 8

The curve C is $y = x^4 - 4x^2$.

(a) Find dy/dx and factorise it completely. (3)

(b) Find all points on C with a horizontal tangent. (3)

(Total for Question 21 is 6 marks)

Curves C_1 and C_2 are $y = x^2 + ax$ and $y = bx - 1$. They touch at the point with $x = 1$.

(a) Find $a - b$. (2)

(b) Given also that the common tangent has gradient 3, find a, b and the point of contact. (3)

(Total for Question 22 is 5 marks)

$C : y = x + \dfrac{4}{x}$, where $x > 0$.

(a) Find the value of x for which the tangent is horizontal. (3)

(b) Find equations of the tangent and normal at this point. (3)

(Total for Question 23 is 6 marks)

The curve C is $y = \dfrac{x^2 + 3x - 4}{\sqrt{x}}$ for $x > 0$.

(a) Express y as a sum of powers of x and find dy/dx. (3)

(b) Find the equation of the tangent at the point on C with $x = 1$. (3)

(Total for Question 24 is 6 marks)

The curve C is $y = x^3 - 3x + 4$. The tangent at P is parallel to the tangent at Q, where $x_P = 2$ and $x_Q \neq 2$.

(a) Find x_Q and the coordinates of P, Q. (3)

(b) Find the y-intercepts of these two tangents. (2)

(Total for Question 25 is 5 marks)

$f(x) = ax^4 + bx^2$. It is given that $f'(1) = 12$ and $f''(1) = 28$.

(a) Find a, b. (4)

(b) Find $f(-1)$ and the gradient of the curve there. (2)

(Total for Question 26 is 6 marks)

The curve C is $y = 3x^2 - 2x + 5$.

(a) Find the point on C where the normal has gradient $-\frac{1}{4}$. (3)

(b) Find the exact area of the triangle formed by this normal and the coordinate axes. (4)

(Total for Question 27 is 7 marks)

A curve has equation $y = x^3 + kx^2$. Its tangent at the point with $x = 1$ passes through $(0, -3)$.

(a) Find k. (3)

(b) For this value of k, find the tangent and the normal at $x = 1$. (4)

(Total for Question 28 is 7 marks)

C is $y = \frac{1}{4}x^4 - 2x^2 + 3x$.

(a) Find dy/dx and d^2y/dx^2. (4)

(b) Find the gradient of the normal at the point where $x = 0$, and give its equation. (3)

(Total for Question 29 is 7 marks)

For $x > 0$, $y = 2x^{5/2} - 5x^{3/2} + 3$.

(a) Find dy/dx and factorise it as $Ax^{1/2}(Bx + C)$. (3)

(b) Find the exact coordinates of the point where the tangent is horizontal. (3)

(Total for Question 30 is 6 marks)

Answers and Suggested Mark Schemes

中文解答与建议评分点

M：方法分；A：准确分，通常依赖对应方法；B：独立得分点。
等价正确解法应获相应分数。评分点为原创建议，并非官方评分细则。

Question 1

P1-C08-R01 · 6 marks

(a) · 4 marks

$6x^2 - 10x + 4$; $12x - 10$.

逐项用幂函数法则，常数导数为0；再对一阶导求导。

M1	一阶幂法则
A1	一阶导
M1	再求导
A1	二阶导

(b) · 2 marks

20.

代入一阶导6+10+4；不是把x代回原曲线。

M1	代入正确导函数
A1	梯度20

Question 2

P1-C08-R02 · 5 marks

(a) · 3 marks

$y = 7x - 6$.

P=(2,8)，导数2x+3在2处为7；y−8=7(x−2)。

B1	正确P
M1	求切线斜率
A1	切线方程

(b) · 2 marks

$(\frac{6}{7}, 0)$ and $(0, -6)$.

对切线方程分别令y=0、x=0。

M1	使用切线而非曲线
A1	两截距

Question 3

P1-C08-R03 · 5 marks

(a) · 2 marks

$2x^{-1/2} - 3x^{-2}$.

将根式写为4x^{1/2}、分式写为3x^{-1}后逐项求导。

M1	写成幂并使用法则
A1	两项正确

(b) · 3 marks

$y = x + 6$.

点(1,7)，切线斜率2−3=−1，法线斜率1。

B1	曲线上点
M1	求切线及负倒数
A1	法线式

Question 4

P1-C08-R04 · 6 marks

(a) · 4 marks

$a+b=3,\ 3a+2b=5;\ a=-1,b=4.$

点条件代原式，梯度条件代导式；消元a=−1,b=4。

M1	正确点条件
M1	正确导数条件
M1	解方程组
A1	两系数

(b) · 2 marks

$-4.$

代系数后二阶导为−6x+8，x=2时−4。

M1	正确二阶导
A1	代值

Question 5

P1-C08-R05 · 5 marks

(a) · 3 marks

$y = 3x^2 - 2 + 5x^{-1}$; $dy/dx = 6x - 5x^{-2}$.

先逐项除x，常数−2导数为0。

M1	正确逐项除法
M1	对化简式求导
A1	完整导式

(b) · 2 marks

Gradient -11; $y = -4$.

导式代−1得−6−5；原化简式给3−2−5。

M1	分别用导式及原式
A1	两值

Question 6

P1-C08-R06 · 5 marks

(a) · 3 marks

$x = \pm\sqrt{2}$.

水平切线要求$3x^2-6=0$，保留正负。

M1	求导并令零
M1	解平方关系
A1	两根

(b) · 2 marks

$y = 2-4\sqrt{2}$ at $x=\sqrt{2}$; $y = 2+4\sqrt{2}$ at $x=-\sqrt{2}$.

在这些点$x^3=2x$，所以$y=2-4x$；配对对应两个x。

M1	正确代入或利用x平方关系
A1	两个配对纵坐标

Question 7

P1-C08-R07 · 7 marks

(a) · 4 marks

Tangent $y = -x/2 + 6$; normal $y = 2x + 1$.

切线用已给斜率；法线斜率为负倒数2，两条都通过P。

M1	切线点斜式
A1	切线
M1	法线斜率
A1	法线

(b) · 3 marks

36 square units.

切线截距$(12,0)$、$(0,6)$，面积$12\times 6/2$。

M1	求切线两截距
M1	正确面积式
A1	面积

Question 8

P1-C08-R08 · 6 marks

(a) · 3 marks

$y = x^3 - x^2 - x - 2$; $dy/dx = 3x^2 - 2x - 1$.

先展开两个因子，再逐项求导。

M1	正确展开
M1	幂法则
A1	导函数

(b) · 3 marks

$(2, 0)$ and $(-\frac{4}{3}, -\frac{130}{27})$.

$3x^2-2x-1=7$给$3x^2-2x-8=0$，$x=2$或$-4/3$；代原式求y。

M1	梯度方程
M1	解两横坐标
A1	两点

Question 9

P1-C08-R09 · 6 marks

(a) · 3 marks

$dh/dx = 0.04x + 0.3$; value 0.7.

按幂法则求导，代x=10。

M1	正确求导
A1	导式
A1	数值

(b) · 3 marks

$0 \le x \le 15$.

0.04x+0.3⩽0.9给x⩽15，再与模型0⩽x⩽20求交。

M1	建立梯度不等式
M1	使用模型定义域
A1	区间

Question 10

P1-C08-R10 · 7 marks

(a) · 2 marks

$\frac{15}{2}x^{1/2} - x^{-1/2}$.

每项系数乘原指数，指数减1。

M1	两项使用分数指数法则
A1	完整导式

(b) At the point where $x = 4$, find

(i) · 3 marks

Point $(4, 36)$; gradient $29/2$.

y=5×8−2×2=36；导数15−1/2=29/2。

B1	点坐标
M1	导式代值
A1	梯度

(ii) · 2 marks

$y = \frac{29}{2}x - 22$.

y−36=(29/2)(x−4)，展开。

M1	正确点斜式
A1	最终方程

Question 11

P1-C08-R11 · 7 marks

(a) · 4 marks

$a = 3, b = -4, c = 3.$

点(0,3)给c=3，切线斜率在0处给b=−4，再由7=4a−8+3求a=3。

B1	c
M1	用切线求b
M1	代另点求a
A1	三系数

(b) · 3 marks

$(2, 7).$

y'=6x−4=8给x=2，再用原曲线求y=7。

M1	梯度匹配
M1	回代原曲线
A1	点

Question 12

P1-C08-R12 · 7 marks

(a) · 4 marks

$1-16x^{-3}$; $48x^{-4}$.

写8x^{−2}再求导两次。

M1	一阶负幂求导
A1	一阶导
M1	再求导
A1	二阶导

(b) · 3 marks

$y=3x+6$.

点(−2,0)，导数1−16/(−8)=3，故y=3(x+2)。

B1	点
M1	求梯度
A1	方程

Question 13

P1-C08-R13 · 6 marks

(a) · 3 marks

$(2, 2)$.

法线斜率1/2意味着切线斜率−2；2x−6=−2给x=2，代回得2。

M1	法切斜率对应
M1	解横坐标
A1	接触点

(b) · 3 marks

$y = \frac{1}{2}x + 1$; $(-2, 0)$.

过(2,2)用斜率1/2，随后令y=0。

M1	正确法线点斜式
A1	方程
A1	截距

Question 14

P1-C08-R14 · 7 marks

(a) · 3 marks

$36x^2 - 48x$.

先得f'=12x^3−24x^2+2，再求导。

M1	正确一阶导
M1	再求导
A1	二阶导

(b) · 4 marks

$x = 0, 4/3$; $f'(0) = 2$, $f'(4/3) = -110/9$.

二阶导12x(3x−4)=0；代入一阶导得到2和−110/9。

M1	解二阶导零点
A1	两x
M1	代一阶导
A1	两梯度

Question 15

P1-C08-R15 · 6 marks

(a) · 3 marks

$p = 2, q = -1.$

3+p=5给p=2；1+p+q=2给q=−1。

M1	正确梯度条件
M1	正确点条件
A1	两参数

(b) · 3 marks

$y = -\frac{1}{5}x + \frac{11}{5}.$

法线斜率−1/5且过(1,2)。

M1	正确法线斜率
M1	点斜式
A1	方程

Question 16

P1-C08-R16 · 6 marks

(a) · 3 marks

$dy/dx = 8x + 4$.

展开$4x^2+4x+1$，再用幂法则。

M1	正确展开
M1	逐项求导
A1	导式

(b) · 3 marks

$(0, 1)$.

给定线斜率$-1/4$，切线应斜率4；$8x+4=4$给$x=0$。

M1	从直线确定所需切线斜率
M1	解梯度条件
A1	点

Question 17

P1-C08-R17 · 7 marks

(a) · 3 marks

$x = \frac{1}{3}, 1.$

$3x^2-4x=-1$给$(3x-1)(x-1)=0$。

M1	求导建立条件
M1	分解求根
A1	两x

(b) · 4 marks

$y = x - 1;\ (0, -1).$

点$(1,0)$，切线斜率-1，法线斜率1。

B1	点
M1	垂线斜率及过点
A1	法线
A1	纵截距

Question 18

P1-C08-R18 · 6 marks

(a) · 3 marks

$dy/dx = -3x^{-3/2} + 2$; value $17/9$.

9^{3/2}=27，故−3/27+2=17/9。

M1	正确负半次幂求导
A1	导式
A1	梯度

(b) · 3 marks

$y = \frac{17}{9}x + 3$.

原曲线y=6/3+18=20；点斜式y−20=(17/9)(x−9)。

B1	正确纵坐标
M1	点斜式
A1	方程

Question 19

P1-C08-R19 · 6 marks

(a) · 3 marks

Tangent is $y=(2a+2)x-a^2$; through the origin gives $a^2=0$.

P=(a,a²+2a)，点斜式整理后的截距为−a²，令其为0。

M1	用a写点与导数
M1	正确切线截距
A1	得a=0

(b) · 3 marks

Tangent $y=2x$; perpendicular $y=-x/2+2$.

a=0时导数2；垂线斜率−1/2，过(2,1)。

B1	切线
M1	垂线斜率及点
A1	垂线方程

Question 20

P1-C08-R20 · 5 marks

(a) · 3 marks

$y = 1$ and $y = 0$, respectively.

导数$6x^2-6x$在0和1都为0；对应纵坐标1和0，所以两条水平切线。

M1	求导并正确代入
M1	求两纵坐标
A1	两水平线

(b) · 2 marks

1.

两条水平直线的垂直距离为纵截距差$|1-0|$。

M1	使用水平线垂直距离
A1	结果

Question 21

P1-C08-R21 · 6 marks

(a) · 3 marks

$4x(x^2-2)=4x(x-\sqrt{2})(x+\sqrt{2})$.

逐项导出$4x^3-8x$，提公因式并平方差分解。

M1	正确求导
M1	提公因式
A1	完全因式分解

(b) · 3 marks

$(0,0)$ and $(\pm\sqrt{2},-4)$.

导数为0的x为0、$\pm\sqrt{2}$，分别代入原式。

M1	导式零点
M1	回代三点
A1	全部坐标

Question 22

P1-C08-R22 · 5 marks

(a) · 2 marks

$a - b = -2$.

接触点具有相同纵坐标：1+a=b−1给a−b=−2；导数相等也给2+a=b。

M1	同点条件
A1	参数差

(b) · 3 marks

$a = 1, b = 3$; contact $(1, 2)$.

切线是C2本身，b=3；差−2给a=1，代点求y=2。

M1	用共同斜率求b
M1	求a与回代
A1	参数及接触点

Question 23

P1-C08-R23 · 6 marks

(a) · 3 marks

$x = 2$.

y′=1−4/x²=0给x²=4，由定义域选2。

M1	正确求导
M1	解并使用定义域
A1	x

(b) · 3 marks

Tangent $y = 4$; normal $x = 2$.

点(2,4)，水平切线的法线是竖直线，不能写成有限斜率的y形式。

B1	正确点
A1	水平切线
A1	竖直法线

Question 24

P1-C08-R24 · 6 marks

(a) · 3 marks

$y = x^{3/2} + 3x^{1/2} - 4x^{-1/2}$; $y\prime = \frac{3}{2}x^{1/2} + \frac{3}{2}x^{-1/2} + 2x^{-3/2}$.

每项除以x^{1/2}，再按幂法则求导；最后一项负系数乘负指数变正。

M1	正确三个幂
M1	分数幂求导
A1	导函数

(b) · 3 marks

$y = 5x - 5$.

点(1,0)，梯度3/2+3/2+2=5。

B1	点
M1	导式代值
A1	方程

Question 25

P1-C08-R25 · 5 marks

(a) · 3 marks

$x_Q = -2$; $P = (2, 6)$, $Q = (-2, 2)$.

导数$3x^2-3$；在2处为9，解$3x^2-3=9$得± 2，排除2。

M1	求P梯度
M1	相等梯度方程
A1	两点

(b) · 2 marks

-12 at P; 20 at Q.

两条线斜率9；截距$y-9x$分别$6-18$与$2+18$。

M1	正确切线截距关系
A1	两个截距

Question 26

P1-C08-R26 · 6 marks

(a) · 4 marks

$a = 2, b = 2$.

4a+2b=12；12a+2b=28；相减8a=16。

M1	一阶条件
M1	二阶条件
M1	消元
A1	两参数

(b) · 2 marks

$f(-1) = 4$; gradient -12.

原式在±1同值，导式$8x^3+4x$在−1给−12。

M1	代入原式和导式
A1	两值

Question 27

P1-C08-R27 · 7 marks

(a) · 3 marks

$(1, 6)$.

切线斜率4，6x−2=4给x=1，代曲线y=6。

M1	由法线求切线斜率
M1	解条件
A1	点

(b) · 4 marks

$\frac{625}{8}$ square units.

法线y=−x/4+25/4，截距25和25/4，面积1/2×25×25/4。

M1	正确法线
M1	求两截距
M1	面积式
A1	精确面积

Question 28

P1-C08-R28 · 7 marks

(a) · 3 marks

$k = 1$.

切点纵坐标1+k、梯度3+2k，切线截距(1+k)−(3+2k)=−2−k；令等于−3。

M1	表示点与梯度
M1	建立截距条件
A1	k

(b) · 4 marks

Tangent $y = 5x - 3$; normal $y = -x/5 + 11/5$.

切点(1,2)，切线斜率5；法线斜率−1/5。

M1	代入参数求点和斜率
A1	切线
M1	法线负倒数
A1	法线

Question 29

P1-C08-R29 · 7 marks

(a) · 4 marks

$x^3 - 4x + 3$; $3x^2 - 4$.

分别逐项求导。

M1	一阶求导
A1	一阶导
M1	二阶求导
A1	二阶导

(b) · 3 marks

Gradient $-\frac{1}{3}$; normal $y = -x/3$.

曲线在原点；切线梯度3，所以法线斜率$-1/3$。

B1	曲线上点
M1	正确负倒数
A1	梯度及方程

Question 30

P1-C08-R30 · 6 marks

(a) · 3 marks

$dy/dx = \frac{5}{2}x^{1/2}(2x-3)$.

导数5x^{3/2}−(15/2)x^{1/2}，提出(5/2)x^{1/2}。

M1	正确求导
M1	提取共同分数幂
A1	要求形式

(b) · 3 marks

$\left(\frac{3}{2}, 3-\frac{3\sqrt{6}}{2}\right)$.

x>0排除x=0；2x−3=0给3/2。原式可写x^{3/2}(2x−5)+3，在3/2处得3−3√6/2。

M1	用定义域找导数零点
M1	精确代入
A1	坐标